Srimanasvi Reddy

ch24btech11039@iith.ac.in | srimanasvireddy@gmail.com | +91 8600162691 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.53
Pioneer Public School	Class XII (CBSE)	2024	94.00%
Hutchings High School	Class X (ICSE)	2022	97.83%

SKILLS

Languages: Python, C, C++, SQL

Machine Learning: Pandas, Matplotlib, NumPy, scikit-learn, Seaborn, XGBoost, Random Forest, SHAP, imbalancedlearn

Tools & Software: Git, GitHub, Jupyter Notebook, Streamlit, Excel, Powerpoint, MATLAB, Canva, Figma

PROJECTS

Credit Risk Modeling for Home Loans

- Developed a credit risk prediction pipeline on the **Home Credit Default Risk** dataset by engineering **20 features** from **3 tables (3.7M+ records)** using SQL aggregations and joins.
- Trained and evaluated multiple models for credit default prediction; addressed **8.1% class imbalance** using class weighting, with **LightGBM achieving ROC-AUC 0.775** on the test set.
- Performed threshold analysis, demonstrating the trade-off between catching high-risk borrowers and minimizing false rejection of low-risk applicants.
- Interpreted model behavior using SHAP, confirming the contribution of engineered financial features.

Resume/Job Description Matcher

- Built a tool using **SentenceTransformer** embeddings for semantic similarity and embedding-based section classification, enabling reliable extraction from documents with inconsistent layouts
- Implemented a three stage skill extraction pipeline combining exact, punctuation-insensitive, and typo-tolerant matching using a custom taxonomy of nearly **200 technical and soft skills**
- Integrated **Llama 3.3 70B via Groq** to generate personalized recommendations based on identified skill gaps, and deployed on **Streamlit** Community Cloud

Predictive Maintenance System for Industrial Equipment

- Addressed severe class imbalance (3.4% failure rate) by comparing SMOTE oversampling against XGBoost's native scale_pos_weight, improving failure **Recall from 13.2% to 83.8%**
- **Engineered 2 features**, improving F1 from 0.683 to 0.717 and **ROC-AUC** from **0.966 to 0.978**
- Deployed an interactive **Streamlit dashboard** with live sensor-input sliders and real-time failure-probability scoring, using SHAP to explain the key drivers behind each prediction.

Hisaab Kitaab-Expense Sharing & Cash Flow Minimization System

- Modelled multi-party debt settlement, minimizing transaction count using a **greedy two-heap approach (O(N log N))** across three split types: equal, selective, and custom per-person amounts.
- Designed a modular object-oriented architecture with constructor overloading and encapsulated split logic.
- Used STL containers and data structures (vector, map, priority_queue) to implement expense tracking, balance computation, and settlement generation.

RELEVANT COURSES & CERTIFICATIONS

CS and AI/ML Courses: Introduction to Programming, Introduction to Artificial Intelligence

Mathematics Courses: Calculus-1&2, Differential Equations, Elementary Linear Algebra, Applied Mathematics for

Chemical Engineering, Numerical Methods

Management Courses: Intro to Entrepreneurship, Communication Skills, Fundamentals of Organizational Structures

Certificates: Supervised Machine Learning: Regression and Classification- Andrew Ng

POSITIONS OF RESPONSIBILITY

Content and Curation Coordinator | TEDx IIT Hyderabad

- Directed content curation and agenda research for **XMUN**, a Model United Nations conference held as a part of TEDx, leading background guide development for 5 committees
- Served as **Vice Chair** for a committee of **15 delegates**, moderating proceedings and debate throughout the conference
- Coordinated speaker content, session flow, and theme ideation for the main event, and co-led scripting, content development, and rehearsal coaching for one of the event's featured speakers

Sponsorship Coordinator | Elan & nVision (Annual Techno-Cultural Fest)

- Worked on sponsorship outreach and coordination
- Assisted in communication with external organizations and internal teams during event planning and execution

EXTRACURRICULAR & COMPETITIONS

- **Amex Strategy Track Campus Case Challenge:** Ranked in the **top 11%** among 17,000+ teams; built a scoring model achieving **88% accuracy** to measure customer profitability on credit card data.
- Completed **NCC B Certificate** and Combined Annual Training Camp (CATC-VII)
- Completed **Bharatanatyam** Level 5 from Akhil Bharatiya Gandharva Mahavidyalaya and Level 3 from Tilak Maharashtra Vidyapeeth